
Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

Anonymous Author(s)

Affiliation

Address

email

Abstract

Biological neurons assign credit through structures that standard neural networks usually ignore: synapses are distributed across branching dendrites, excitatory and inhibitory inputs evoke synaptic conductances, and inhibition can shunt local voltage by changing total input resistance. We study conductance-based dendritic networks with E/I synapse banks, shunting inhibition, and tree-structured branch-to-soma coupling, and ask when a low-bandwidth somatic broadcast can approximate compartment-specific backpropagated errors. Exact gradients for these dendritic trees have the form $\text{gradient} = \text{local eligibility} \times \text{compartment error}$: the eligibility term uses presynaptic activity, driving force, and input resistance, while the sole fast non-local term is a path-specific compartment error. Under identity upward transfer, that error is the somatic error transported through conductance-stage path gains. This factorization turns local learning into a credit-signal compression problem. Shunting is useful when it reshapes the path-specific error field so that a low-rank broadcast is a better approximation, not because it makes LocalCA a close copy of backpropagation. We test this mechanism across exact-gradient reconstruction, oracle-transport, inhibition-intervention, competence, morphology, and harder-data diagnostics. Shunting reduces gradient-scale mismatch and makes exact compartment errors more rank-1 compressible than matched additive controls in the regimes studied here. Inhibitory-conductance interventions show that learned inhibition carries sample-specific pathway structure, while transported-error oracle experiments show that restoring the exact effective path-gain error substantially reduces the local-learning gap. Under nonnegative conductances and rank-1 broadcast, 5F shunting LocalCA remains 5–6 percentage points below matched backpropagation on MNIST, Fashion-MNIST, and context gating. These results suggest that E/I conductance, shunting inhibition, and dendritic branching can reshape credit-signal geometry and make low-bandwidth local learning more faithful in the regimes studied.

1 Introduction

Unlike point units in standard neural networks, biological neurons assign credit across branching dendritic trees. A synapse does not only see presynaptic activity and a scalar postsynaptic activation; it also has access to local membrane voltage, synaptic reversal potential, total conductance, and input resistance. Inhibitory synapses can therefore do more than subtract activity: by increasing total local conductance, inhibition can shunt voltage and change the gain of every dendritic path passing through that compartment.

We examine whether these biophysical ingredients can facilitate local credit assignment (LocalCA). Backpropagation [15] would assign a distinct error to every dendritic compartment, whereas a biologically plausible supervisory signal is more likely to arrive as a low-bandwidth somatic or modulatory broadcast [20]. The central question is therefore not whether such a broadcast can

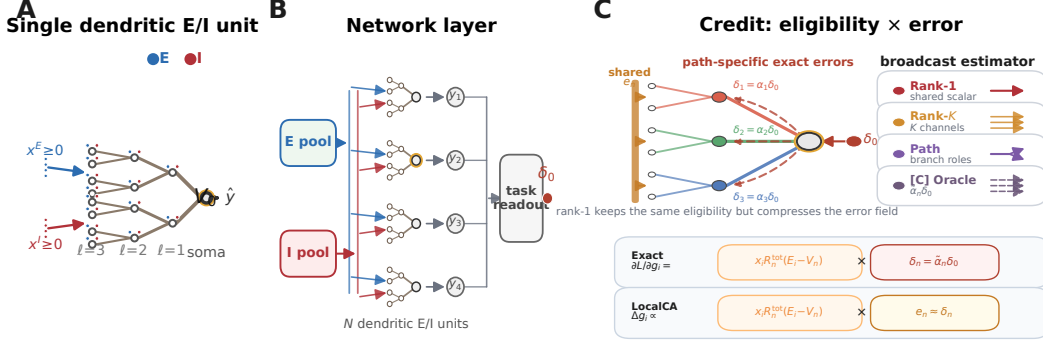


Figure 1: **Model and credit assignment.** (A) Single dendritic E/I unit with nonnegative excitatory and inhibitory input streams, branch synapse banks, and a gold-ringed soma matching the highlighted unit in B. (B) Layer of dendritic E/I units projecting to a task readout; $\delta_0 = \partial L / \partial V_0$ is the somatic error available for broadcast. (C) Credit factorization: exact updates use the same local eligibility multiplied by path-specific compartment errors δ_n , whereas LocalCA keeps the eligibility and replaces δ_n with a broadcast estimator e_n ; Rank-1, Rank-K, Path, and Oracle broadcasts trade bandwidth for fidelity.

reproduce unconstrained backpropagation in every setting, but when the exact dendritic error field is simple enough for it to approximate. Our contribution is not simply to add dendritic structure to a neural network, but to show how conductance-based synapses, shunting inhibition, and dendritic topology change the geometry of the credit signal itself.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single non-local compartment error:

$$\text{gradient} = \underbrace{\text{local eligibility}}_{\text{presynaptic drive, } E-V, R^{\text{tot}}} \times \underbrace{\text{compartment error}}_{\text{path-specific non-local term}}.$$

The compartment error is path-specific: under identity upward transfer it is the somatic error multiplied by a conductance-stage path gain through the dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

What is local in LocalCA? The synapse-local factor is the eligibility term: presynaptic drive, local voltage, driving force, input resistance, and optional reliability gates such as $r_n^{\text{4F}} \phi_n$. The fast task-dependent non-local quantity is the compartment error, approximated by a broadcast field e_n . Thus LocalCA variants differ by the broadcast estimator and optional reliability gates, while preserving the same local eligibility mechanism.

This factorization turns local credit assignment into a compressibility problem. By compressibility, we mean that the matrix of exact compartment errors across examples and compartments is well approximated by a low-rank signal, especially a rank-1 field shared across compartments. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help monotonically. It helps rank-1 broadcast when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field more compressible.

Related work. This work builds on models of dendritic signal integration, branch-level nonlinear computation, divisive normalization, and biologically plausible teaching signals [1, 3, 4, 8, 9, 17–19, 22, 26–28, 34, 35, 45]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [13–16, 21, 29–33, 36–40]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically. Gradient-fidelity diagnostics are used as diagnostics, not as the main claim of backprop equivalence: shunting has a higher final LocalCA to backprop cosine than the additive control, but the cosine remains modest, while scale mismatch is substantially reduced. Exact-error compressibility and

inhibition-intervention diagnostics show that learned inhibitory conductance carries sample-specific pathway structure. Transported-error oracle experiments show that supplying the exact effective path-gain-transported compartment error substantially reduces the local-learning gap, identifying broadcast fidelity as an important bottleneck. In an MNIST gradient-dynamics diagnostic, final cosine is 0.100 ± 0.033 for shunting vs. 0.005 ± 0.026 for additive, and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05 . On noise resilience under matched inhibition ($N_I \geq 5$), additive and shunting LocalCA are close under the practical broadcast, while supplying the [C] transported-error oracle lifts both substantially. This isolates broadcast fidelity, rather than the eligibility rule alone, as the bottleneck.

Contributions. This paper makes three contributions: deriving the conductance-stage path gain, its local-eligibility/path-error factorization, and a closed-form inhibitory path-gain ratio; identifying path-gain compressibility as the quantity mediating whether low-rank somatic broadcast can approximate exact compartment errors; and showing empirically that shunting inhibition can improve this approximation while also identifying regimes where rank-1 feedback fails.

Claim status. We distinguish [A] analytical claims (factorization, path gains, inhibitory path-gain derivatives), [C] codebase/simulator diagnostics (exact reconstruction, broadcast comparisons, transported-oracle upper bounds, additive/shunting controls), and [B] biological predictions (branch-local inhibition and morphology-dependent credit routing). Transported-error results are always [C] oracle diagnostics, not proposed biological learning rules.

2 Compartmental Voltage Model and Gradient Derivation

For each dendritic compartment, we use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal potentials are set to 0, the excitatory reversal potential is set to 1, and leak conductance is fixed at 1 (i.e., all conductances are expressed relative to leak). Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force ($E - V$) and total input resistance R^{tot} . Throughout the paper we write dendritic morphologies as rooted trees $[b_1, b_2, \dots, b_D]$, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus [3, 3] means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance-based synapse banks and, when enabled, learned inhibitory branch-level conductance-based synapse banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level synaptic integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in path gain through input resistance. The local learning rule then asks whether a low-bandwidth teaching signal can stand in for the exact path-specific error induced by that tree. **Notation.** We use α_n^{cond} for the conductance-stage path gain from compartment n to the soma and $\tilde{\alpha}_n$ for the effective path gain after including any post-voltage activation derivatives; when upward transfer is identity, $\tilde{\alpha}_n = \alpha_n^{\text{cond}}$. We use N_E and N_I for the number of excitatory and inhibitory synapses per branch, r_n^{4F} for the empirical 4F branch-soma covariance proxy, ϕ_n for the 5F confidence factor, λ_{mix} for the pathway-broadcast mixing weight, and κ_j for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

Voltage equation and local sensitivities. Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance

122 $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

123 V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup$
 124 $\{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow
 125 directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

126 Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage
 127 difference, driving force, and input resistance.

128 *Shunting inhibition as divisive gain control.* An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current
 129 $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds
 130 to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but
 131 its effect on firing rates can be subtractive in some regimes [10]. Inhibitory plasticity can balance
 132 excitation dynamically [11]; our learned inhibitory conductances serve an analogous role. The same
 133 denominator also gives the direct connection to path gain. Writing $G_n^E(x)$ and $G_n^I(x)$ for the total
 134 excitatory and inhibitory synaptic conductance impinging on compartment n ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

135 Increasing $G_n^I(x)$ lowers $R_n^{\text{tot}}(x)$ and therefore multiplicatively suppresses every upstream path
 136 whose error must pass through compartment n . In this sense inhibition can gate credit flow by
 137 changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2
 138 formalizes this statement after the tree path gain is defined.

139 *Exact gradients for dendritic trees.* Let V_0 be the somatic output and define the soma/core error
 140 as $\delta_0 := \partial L / \partial V_0$. For a linear decoder $\hat{y} = W_{\text{dec}} V_0$, this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a
 141 nonlinear decoder, δ_0 is the corresponding decoder-input Jacobian product. The theoretical derivation
 142 is stated for the conductance-stage voltage before any post-voltage reactivation. When a branch
 143 applies a monotone reactivation before transmitting activity upward, the same recursion holds with
 144 additional local derivative factors along the path. All exact-error diagnostics below are computed as
 145 pre-activation voltage errors, including activation-derivative factors when reactivation is enabled.

146 **Theorem 1** (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree*
 147 *with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, suppose the parent*
 148 *receives either the pre-activation voltage V_n or a local post-voltage activity $a_n = f_n(V_n)$. For*
 149 *identity upward transfer, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

150 *with boundary condition $\partial L / \partial V_0 = \delta_0$. With post-voltage transfer $a_n = f_n(V_n)$, the local factor*
 151 *becomes $f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$. Because the graph is a tree, each compartment has a unique path to*
 152 *the soma. Defining the conductance-stage path gain*

$$\alpha_n^{\text{cond}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n^{\text{cond}} \delta_0, \quad (4)$$

153 *with $\alpha_0^{\text{cond}} = 1$ under identity transfer. In the post-voltage case, the exact effective path gain is*

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \tilde{\alpha}_n \delta_0, \quad (5)$$

154 *where $f'_i(V_i)$ is omitted for identity transfer.*

155 *Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1. $\square$

156 The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer
157 from the non-local somatic error.

158 **Proposition 2** (Inhibitory control of conductance-stage path gain). *Let $\mathcal{A}(n)$ be the set of com-*
159 *partments on the path from compartment n to the soma, excluding n and including the parent*
160 *compartments through which the error must pass. Holding dendritic coupling conductances fixed, an*
161 *added inhibitory conductance $\Delta G_k^I(x) \geq 0$ at any $k \in \mathcal{A}(n)$ changes the conductance-stage path*
162 *gain by*

$$\frac{\alpha_n^{\text{cond}}(x; \Delta G^I)}{\alpha_n^{\text{cond}}(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (6)$$

163 *Proof.* Substitute $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$ into the path-gain product in Eq. (4). Adding inhibitory conductance
164 changes only the denominator at compartments on the path from n to the soma, which gives the
165 product ratio and the log derivative in Eq. (6). $\square$

166 Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths
167 below the inhibited compartment. Second, inhibition is useful for rank-1 local learning only when
168 this attenuation makes the distribution of α_n^{cond} more uniform or better aligned with the task. If
169 inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population,
170 it can suppress useful signals and hurt learning. In the additive control the same inhibitory input
171 changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as
172 a multiplicative resistance gate.

173 **When does shunting compress path gains?** The attenuation identity above is not by itself a
174 concentration result. Let $z_n = \log \alpha_n^{\text{cond}}$ be the log conductance-stage path gain for compartment n .
175 If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left(1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (7)$$

176 then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2 \text{Cov}(z, \beta). \quad (8)$$

177 Thus shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (9)$$

178 Eq. (7) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (8) then applies the
179 variance identity for $z - \beta$. Eq. (9) is a diagnostic condition under one averaging measure, not the
180 empirical mechanism claimed below. In finite trained networks, the covariance margin can disagree
181 with CV and exact-error-rank summaries because these statistics average over different compartment
182 and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and
183 inhibition-intervention measurements as the empirical tests, and report the covariance margin as a
184 boundary check in the appendix.

185 **Corollary 1** (Local–Global Factorization). *The exact synaptic gradient at compartment n factorizes*
186 *as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (10)$$

187 *where the eligibility term depends only on quantities available at synapse i (presynaptic activity*
188 *x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole*
189 *non-local quantity.*

190 The factorization in Eq. (10) isolates the fast task-error-dependent non-local quantity as $\partial L / \partial V_n =$
191 $\alpha_n^{\text{cond}} \delta_0$ under identity upward transfer and $\partial L / \partial V_n = \tilde{\alpha}_n \delta_0$ when post-voltage transfer is enabled.

The 4F and 5F factors introduced below use slower voltage statistics, so they are empirical reliability preconditioners rather than additional error signals. Any practical local rule therefore depends on how well its broadcast signal approximates that exact compartment error, which we evaluate in the gradient-fidelity diagnostic below. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot} , which narrows the spread of local sensitivities across the tree. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation.

3 Local Learning Rules

Broadcast error approximation. We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0 . For a linear readout this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L/\partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top (\partial L/\partial \hat{y})$ so that the broadcast remains defined in soma/core coordinates. In all main rank-1 experiments, this soma/core error is deliberately compressed to one scalar per example, $\bar{\delta}_b = d_0^{-1} \sum_{c=1}^{d_0} \delta_{0,bc}$, and broadcast to all compartments of that example. For higher-bandwidth controls, P_K is a fixed random projection from soma-error coordinates to K feedback channels, $q_{n,k}$ are fixed compartment-specific mixing coefficients, and Γ_n denotes the pathway-structured transport map for compartment n . We distinguish five feedback objects: **rank-1 shared broadcast**, $e_n = \bar{\delta} \mathbf{1}$; **neuron-wise/per-soma broadcast**, $e_n = \delta_0$ when the experiment explicitly keeps the soma-coordinate error vector; **rank- K random broadcast**, $c = P_K \delta_0$ and $e_n = \sum_k q_{n,k} c_k$; **path-structured broadcast**, $e_n = \lambda_{\text{mix}} \bar{\delta} \mathbf{1} + (1 - \lambda_{\text{mix}}) \Gamma_n(\delta_0)$ with pathway roles motivated by vectorized dendritic teaching signals [44]; and the **transported oracle**, $e_n^{\text{transport}} = \tilde{\alpha}_n \delta_0 = \partial L/\partial V_n$, an analysis upper bound because it supplies the effective path gain from Theorem 1. Operationally, path-transport runs with reactivation enabled transport activation-space errors through parent reactivation slopes and multiply by the local $f'_n(V_n)$ before conductance updates; with identity transfer this reduces to $\alpha_n^{\text{cond}} \delta_0$. Thus the main feedforward result tests an explicit low-bandwidth compression of decoder-input/soma error, not an output-dimensional fallback (Appendix Table S14). Under shunting, this deliberately low-bandwidth field supports nontrivial learning on the standard classification tasks, while routed tasks and rank-bridge controls in the appendix expose regimes where rank-1 broadcast collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

Three-factor learning rule (3F).

Definition 1 (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (11)$$

where $\langle \cdot \rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force ($E_j - V_n$). We write Δg as the local gradient estimate supplied to the optimizer; the optimizer applies the usual descent step. Equivalently, one could absorb the minus sign into e_n and treat it as a descent signal.

Additive control. Rule (11) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive E/I control, inhibition enters as a fixed signed voltage contribution rather than through the conductance denominator:

$$V_n^{\text{add}} = \sum_{j \in E} g_j^E x_j^E - \sum_{j \in I} g_j^I x_j^I + \sum_k g_k^{\text{den}} V_k.$$

Equivalently, $\partial V_n^{\text{add}} / \partial g_j^{\text{syn}} = s_j x_j$ with $s_j = +1$ for excitatory and $s_j = -1$ for inhibitory synapses, and $\partial V_n^{\text{add}} / \partial g_k^{\text{den}} = V_k$. The additive local gradient therefore has no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model. In implementations

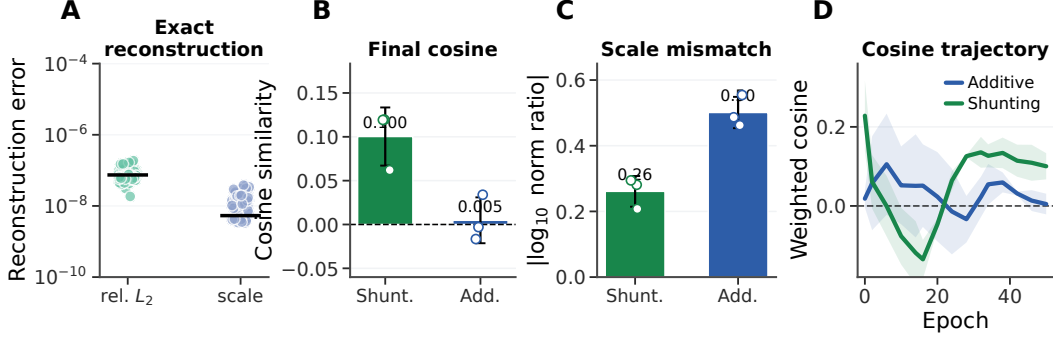


Figure 2: **Exact reconstruction and gradient diagnostics.** (A) Exact-factorization reconstruction matches autograd to numerical precision. (B) Final weighted LocalCA to backprop cosine, reported as a diagnostic rather than the main mechanism claim; dots show seeds. (C) Final scale mismatch, $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$; dots show seeds. (D) Cosine trajectory over learning; Appendix Fig. S4 shows finite gradient norms. Error bars are ± 1 s.d. across 3 seeds.

with unconstrained raw parameters θ and $g = \text{softplus}(\theta)$, Eqs. (11)–(12) give conductance-space gradients; before assigning gradients to θ , we multiply by $\partial g/\partial \theta$. The exact-gradient reconstruction diagnostics include this parameterization factor. The compact mechanism table below summarizes the matched shunting/additive comparison.

Property	Shunting / conductance core	Additive control
Voltage	Normalized by total conductance	Signed E/I voltage sum
Inhibition	Divisive conductance load	Subtractive signed term
Synaptic eligibility	$xR^{\text{tot}}(E - V)e$	$sxe, s \in \{+1, -1\}$
Driving force	Present through $E - V$	Absent from additive derivative
Input resistance	State-dependent, $R^{\text{tot}} = (g^{\text{tot}})^{-1}$	Effectively fixed
Credit interpretation	Inhibition changes path gain and voltage sensitivity	Inhibition shifts voltage but does not gate path gain

Practical heuristic wrappers: 4F and 5F. 4F (branch-soma relevance proxy). In the conductance-stage recursion (4), the path gain α_n^{cond} attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$r_n^{4F} = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

Positive r_n^{4F} up-weights branches whose voltages covary with the soma; weak or sign-opposed covariance reduces the reliability of the shared broadcast for that branch. We keep 4F as a diagnostic intermediate rather than a theorem-derived path-gain estimate. Empirically, this factor alone gives little improvement over 3F; the practical jump comes from the 5F confidence factor below.

5F (confidence-weighted 4F). 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally, ϕ_n is a branch-wise confidence gate, clamped to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived; its gain should be read as practical denoising rather than theoretical necessity. It is positive and bounded, and rescales locally estimated reliability from compartment and parent-voltage statistics rather than introducing a new error signal.

Definition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta r_n^{4F} \phi_n \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B. \quad (12)$$

Eq. (12) is therefore the main practical LocalCA update. The 3F rule uses the theorem-derived local eligibility with an approximate broadcast error, while 4F and 5F add empirical reliability factors. A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

4 Experiments

Setup. The experiments test five linked predictions: exact factorization should reconstruct autograd gradients; shunting should improve the scale and structure of LocalCA credit signals even when

directional cosine is weak; exact compartment errors should be more compressible under shunting than under matched additive controls; learned inhibition should carry sample-specific pathway structure rather than merely adding conductance load; and replacing the broadcast with effective transported error should substantially reduce the local-learning gap. We evaluate capacity-calibrated supervised tasks and controlled sweeps that isolate inhibition strength, rule family, broadcast mode, morphology, and architecture. The main text uses MNIST [42], Fashion-MNIST [41], context gating, noise resilience, and a compressed morphology-regime summary; CIFAR-10 [43] and routed-feedback diagnostics are reported in the appendix (Figs. S8, S12). Noise resilience is a nonnegative MNIST corruption task with a fixed latent Gaussian perturbation projected into input space and clipped to $[0, 1]$. Context gating is a binary contextual figure-ground MNIST task in which the digit half and context/noise half must be integrated under the nonnegative-drive constraint. Main comparisons are architecture-matched additive vs. shunting dendritic cores and local vs. backprop training; DFA is a supplementary baseline. Unless otherwise stated, local runs use 5F with explicit per-example scalar rank-1 compression of the soma/core error; vector-valued, low-rank, pathway-structured, and transported-error controls are labeled separately. All main-claim dendritic runs use nonnegative conductance transforms and nonnegative first-layer synaptic drive; image inputs are in $[0, 1]$, corrupted inputs are clipped back to $[0, 1]$, and context gating uses a nonnegative contextual figure-ground MNIST construction. For context gating, the main LocalCA competence runs use an HSIC auxiliary term with weight 0.01; the appendix reports the HSIC ablation. The transported-error condition is an oracle upper bound derived from Theorem 1, not the proposed biological rule. Main mechanistic, oracle, and competence results use 5 seeds per condition unless noted otherwise; Appendix A defines the weighted gradient, path-gain CV, and exact-error SVD metrics.

From exact factorization to credit-signal fidelity.

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Across the MNIST gradient-dynamics diagnostic, shunting is closer to backprop than the additive control at the final checkpoint, but the directional agreement is still weak in absolute terms: weighted cosine rises from 0.005 ± 0.026 to 0.100 ± 0.033 , and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05 . Figure 2A confirms that the implementation matches the theory under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics in the figure. Figure 2D shows the same gradient-alignment comparison over training: cosine is a noisy diagnostic and can be transiently weak, while additive alignment remains near zero at the final checkpoint rather than failing only at one static snapshot. Appendix Fig. S4 verifies that the corresponding local and backprop gradient norms remain finite throughout learning, ruling out a trivial zero-gradient explanation. At the level of the 3F rule, the central non-local approximation is therefore the broadcast substitute for $\partial L/\partial V_n$.

Mechanistic chain: path gains, broadcast fidelity, and oracle transport. Figure 3 provides the central empirical chain. At matched inhibition ($N_I=5$), shunting narrows the MNIST conductance-stage path-gain field (CV 0.23 vs. 1.16 for additive) and makes the exact compartment-error matrix more compressible: the best rank-1 residual drops from 0.83 ± 0.07 to 0.65 ± 0.18 , while participation rank drops from 8.5 ± 3.3 to 3.2 ± 1.2 . This is the diagnostic matched to the paper’s main claim: rank-1 broadcast is useful when the exact error field is closer to low rank.

The inhibition intervention shows that the trained computation depends on the learned inhibitory field, rather than only on a matched average conductance load: zeroing, shuffling, or clamping learned branch inhibition drops MNIST from about 90% to 20–26% and noise resilience from about 81% to 10–19%. The fidelity and oracle panels probe the same broadcast bottleneck: rank-1 broadcast tracks the exact compartment error better under shunting once inhibition is present, transported error reaches about 0.95–1.0 fidelity, and transported learning lifts both the shunting and additive networks to 92–95% on noise resilience. CIFAR-10 and direct-I versus explicit-I path-gain probes remain appendix diagnostics (Fig. S8; Table S7).

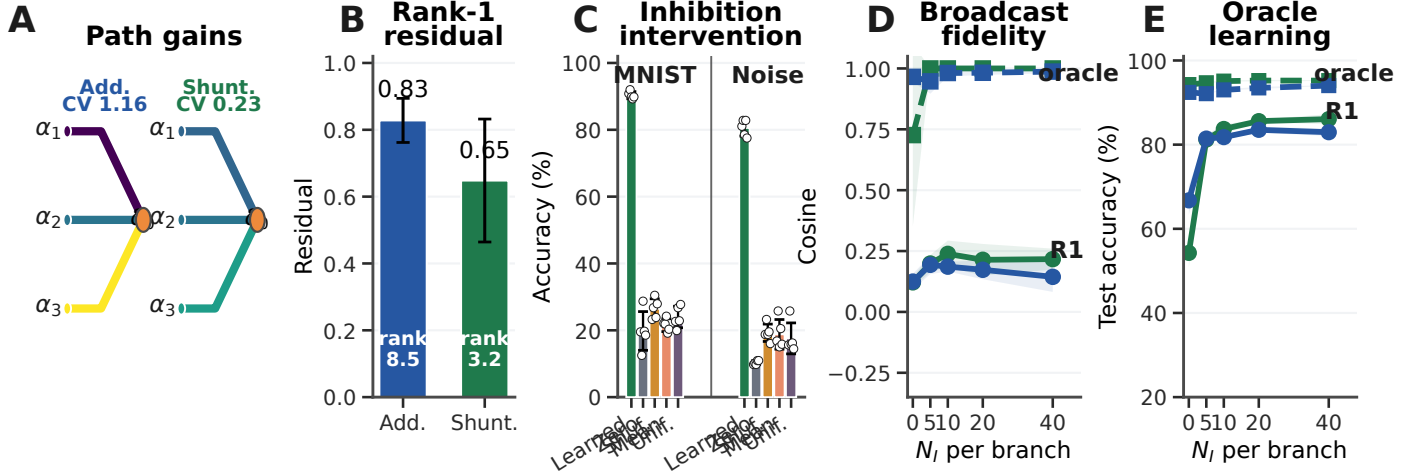


Figure 3: **Mechanistic chain from path gains to learning.** (A) Path-gain fields at matched inhibition. (B) Exact-error compressibility on MNIST. (C) Post-training inhibitory-conductance interventions (learned, zero, shuffled, mean-clamped, or uniform matched inhibition). (D) Compartment-error fidelity on noise resilience; dashed lines show the [C] transported-error oracle. (E) Transported-error learning where rank-1 broadcast fails. Error bars are ± 1 s.d.

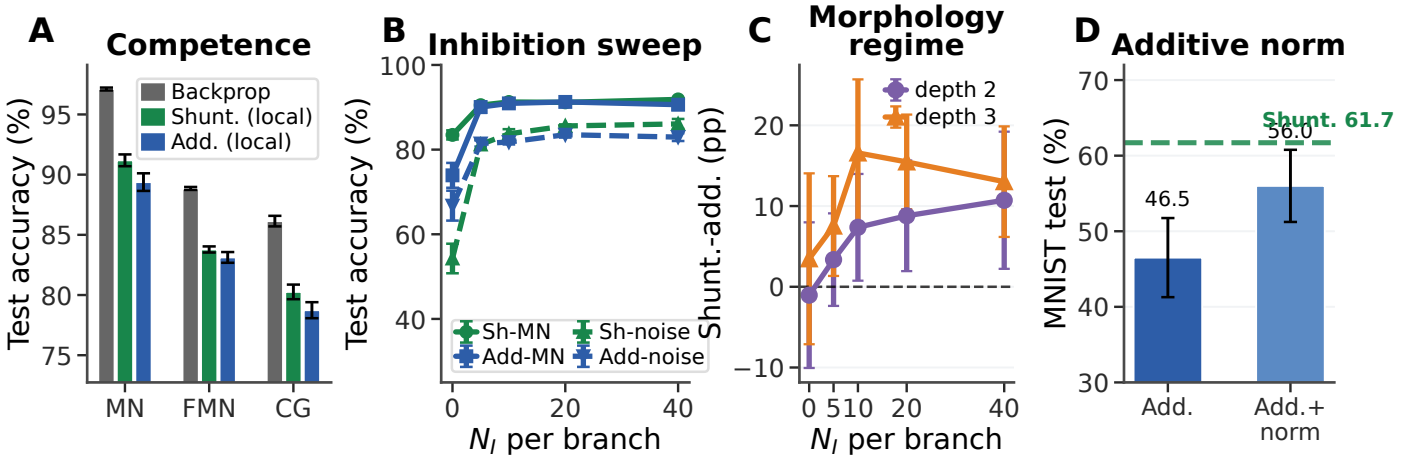


Figure 4: **Capacity-calibrated competence and regime dependence.** (A) Matched backprop vs. rank-1 5F LocalCA on MNIST, Fashion-MNIST, and context gating. (B) Inhibitory dose-response. (C) Morphology-dependent shunting advantage on noise resilience. (D) Separate stress-regime additive-normalization diagnostic. Error bars denote ± 1 s.d. where available.

320 *Capacity-calibrated competence.* In the capacity-calibrated regime, shunting LocalCA remains within
 321 5–6 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and context gating
 322 (Table S5; Fig. 4). In a separate lower-capacity stress-regime diagnostic, Fig. 4D shows that additive
 323 normalization improves additive LocalCA but remains below the shunting reference, so the gain is
 324 not explained by generic voltage normalization alone. Appendix Fig. S1 summarizes the rule and
 325 feedback controls: 5F is the strongest practical optimizer, somatic rank-1 error is essential, and the
 326 noise-resilience rank bridge improves as feedback bandwidth increases from low-rank $K=1$ to $K=8$,
 327 with [C] path transport highest.

328 *Regime dependence of the shunting advantage.* With matched rank-1 broadcast, shunting is modestly
 329 better on MNIST, Fashion-MNIST, and context gating, including 0.803 ± 0.006 vs. 0.787 ± 0.007
 330 for additive on context gating. The gap grows on inhibition-sensitive noise resilience once inhibitory
 331 conductance is present (Fig. 4B). Morphology shifts the useful inhibitory operating regime: different
 332 tree depths and branch factors peak at different N_I , and the shunting advantage is not monotonic in

inhibition (Fig. 4C). Appendix Fig. S6 extends this boundary check to depth scaling and noisy error broadcasts.

5 Discussion and Limitations

We studied local credit assignment in conductance-based dendritic networks with explicit E/I conductance synapses, shunting inhibition, and tree-structured branching. Exact gradients separate synapse-local eligibility–presynaptic activity, driving force, and input resistance—from one path-specific compartment error, making LocalCA an error-approximation problem.

The experiments support a coherent mechanism. Branching creates credit paths, E/I synapse banks determine the conductance state in each compartment, and shunting modulates input resistance along those paths. In the regimes studied here, this makes exact errors more rank-1 compressible and reduces gradient-scale mismatch; LocalCA to backprop cosine improves relative to additive controls but remains a modest diagnostic rather than the central evidence. Transported-error oracles substantially reduce the local-learning gap, while inhibition interventions show that learned inhibitory conductance carries sample-specific pathway structure rather than only global conductance scale. Appendix Table S8 states direct biological predictions of the same path-gain mechanism.

The limits are important. Shunting is not universally beneficial, and more inhibition is not always better. The practical 5F rule is empirical; Eq. (9) is interpretive; activation shape, morphology, task structure, and broadcast rank all change the operating regime. We do not claim competitive CIFAR-10 performance, a recurrent inhibitory circuit model, or that scalar broadcast suffices for routed/pathway-specific tasks.

References

- [1] Koch, C. (1999). *Biophysics of Computation*. Oxford University Press.
- [2] Dayan, P., & Abbott, L. F. (2001). *Theoretical Neuroscience*. MIT Press.
- [3] Poirazi, P., Brannon, T., & Mel, B. W. (2003). Pyramidal neuron as two-layer neural network. *Neuron*, 37(6), 989–999.
- [4] London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28, 503–532.
- [5] Spruston, N. (2008). Pyramidal neurons: dendritic structure and synaptic integration. *Nature Reviews Neuroscience*, 9, 206–221. <https://doi.org/10.1038/nnr2286>
- [6] Branco, T., & Häusser, M. (2010). The single dendritic branch as a fundamental functional unit in the nervous system. *Current Opinion in Neurobiology*, 20(4), 494–502. <https://doi.org/10.1016/j.conb.2010.07.009>
- [7] Larkum, M. (2013). A cellular mechanism for cortical associations: an organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3), 141–151. <https://doi.org/10.1016/j.tins.2012.11.006>
- [8] Urbanczik, R., & Senn, W. (2014). Learning by the dendritic prediction of somatic spiking. *Neuron*, 81(3), 521–528.
- [9] Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13(1), 51–62.
- [10] Holt, G. R., & Koch, C. (1997). Shunting inhibition does not have a divisive effect on firing rates. *Neural Computation*, 9(5), 1001–1013.
- [11] Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, 334(6062), 1569–1573.

- [12] Hennequin, G., Agnes, E. J., & Vogels, T. P. (2017). Inhibitory plasticity: balance, control, and codependence. *Annual Review of Neuroscience*, 40, 557–579. <https://doi.org/10.1146/annurev-neuro-072116-031005>
- [13] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276.
- [14] Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J., & Hinton, G. (2020). Backpropagation and the brain. *Nature Reviews Neuroscience*, 21(6), 335–346. <https://doi.org/10.1038/s41583-020-0277-3>
- [15] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323, 533–536.
- [16] Nøklund, A. (2016). Direct feedback alignment provides learning in deep neural networks. *NeurIPS*, 29.
- [17] Guerguiev, J., Lillicrap, T. P., & Richards, B. A. (2017). Towards deep learning with segregated dendrites. *eLife*, 6, e22901.
- [18] Sacramento, J., Costa, R. P., Bengio, Y., & Senn, W. (2018). Dendritic cortical microcircuits approximate the backpropagation algorithm. *NeurIPS*, 31.
- [19] Whittington, J. C., & Bogacz, R. (2019). Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3), 235–250.
- [20] Richards, B. A., & Lillicrap, T. P. (2019). Dendritic solutions to the credit assignment problem. *Current Opinion in Neurobiology*, 54, 28–36.
- [21] Scellier, B., & Bengio, Y. (2017). Equilibrium propagation. *Frontiers in Computational Neuroscience*, 11, 24.
- [22] Song, Y., Millidge, B., Salvatori, T., Lukasiewicz, T., Xu, Z., & Bogacz, R. (2024). Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27, 348–358. <https://doi.org/10.1038/s41593-023-01514-1>
- [23] Gretton, A., Bousquet, O., Smola, A. J., & Schölkopf, B. (2005). Measuring statistical dependence with Hilbert-Schmidt norms. *Algorithmic Learning Theory*, 63–77.
- [24] Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3), 419–420.
- [25] Turrigiano, G. G. (2008). The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435.
- [26] Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019.
- [27] Greedy, W., Zhu, H. W., Pemberton, J., Mellor, J., & Ponte Costa, R. (2022). Single-phase deep learning in cortico-cortical networks. *NeurIPS*, 35, 24213–24225.
- [28] Haider, P., Ellenberger, B., Kriener, L., Jordan, J., Senn, W., & Petrovici, M. A. (2021). Latent Equilibrium: A unified learning theory for arbitrarily fast computation with arbitrarily slow neurons. *NeurIPS*, 34.
- [29] Hinton, G. (2022). The forward-forward algorithm. *arXiv:2212.13345*.
- [30] Dellaferrera, G., & Kreiman, G. (2022). Error-driven input modulation: Solving the credit assignment problem without a backward pass. *Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research*, 162, 4937–4955.
- [31] Lee, D.-H., et al. (2015). Difference target propagation. *ECML*, 498–515.

- 422 [32] Meulemans, A., et al. (2021). Credit assignment in neural networks through deep feedback
423 control. *NeurIPS*, 34.
- 424 [33] Millidge, B., Seth, A. K., & Buckley, C. L. (2021). Predictive coding: A theoretical and
425 experimental review. *arXiv:2107.12979*.
- 426 [34] Koch, C., Poggio, T., & Torre, V. (1983). Nonlinear interactions in a dendritic tree. *PNAS*, 80(9),
427 2799–2802.
- 428 [35] Silver, R. A. (2010). Neuronal arithmetic. *Nature Reviews Neuroscience*, 11(7), 474–489.
- 429 [36] Max, K., Kriener, L., Pineda García, G. et al. (2024). Learning efficient backprojec-
430 tions across cortical hierarchies in real time. *Nature Machine Intelligence*, 6, 619–630.
431 <https://doi.org/10.1038/s42256-024-00845-3>
- 432 [37] Ma, C., Wu, J., Si, C., & Tan, K. C. (2024). Scaling supervised local learning with augmented
433 auxiliary networks. *International Conference on Learning Representations (ICLR)*.
- 434 [38] Lv, C., Xu, J., Lu, Y., Wang, X., Wang, Z., Xu, Z., Yu, D., Du, X., Zheng, X., & Huang, X.
435 (2025). Dendritic Localized Learning: Toward Biologically Plausible Algorithm. *Proceedings*
436 *of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning*
437 *Research*, 267, 41682–41700.
- 438 [39] Erdogan, M., Pehlevan, C., & Erdogan, A. T. (2025). Error Broadcast and Decorrelation
439 as a Potential Artificial and Natural Learning Mechanism. *Advances in Neural Information*
440 *Processing Systems*, 38.
- 441 [40] Kao, C.-H., & Hariharan, B. (2024). Counter-Current Learning: A Biologically Plausible Dual
442 Network Approach for Deep Learning. *Advances in Neural Information Processing Systems*,
443 37.
- 444 [41] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for bench-
445 marking machine learning algorithms. *arXiv:1708.07747*.
- 446 [42] LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to
447 document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324.
- 448 [43] Krizhevsky, A. (2009). Learning multiple layers of features from tiny images. Technical report,
449 University of Toronto.
- 450 [44] Francioni, V., Tang, V. D., Toloza, E. H. S. et al. (2026). Vectorized instructive signals in cortical
451 dendrites. *Nature*, 652, 1254–1263. <https://doi.org/10.1038/s41586-026-10190-7>
- 452 [45] Iyer, A., Grewal, K., Velu, A., Souza, L. O., Forest, J., & Ahmad, S. (2022). Avoiding catas-
453 trophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in*
454 *Neurorobotics*, 16, 846219.

Diagnostic	Regime	Cached source	Use in paper
Gradient / N_I sweep	Nonnegative inputs, [128, 128] E layers, [3, 3] tree, 5F rank-1, 5 seeds	analysis/gradient_fidelity_vs_ie_nonnegativeinput_fix/gradient_fidelity_summary.csv	Active Fig. 4B and supplementary N_I details
Theory diagnostics	Same nonnegative-input sweep, hooks for exact errors and path gains	analysis/theory_diag_gradient_fidelity_vs_ie_nonnegativeinput_fix_summary/theory_diag_by_condition.csv	Active path-gain, fidelity, and exact-reconstruction diagnostics
Rank bridge	Noise resilience, shunting, low-rank $K=1, 2, 4, 8$ and path transport, 5 seeds	analysis/rank_bridge_nonnegativeinput_fix/noise_resilience_rank_bridge_summary.csv	Appendix Fig. S1C and Table S9
Broadcast-noise stress	Separate error-noise stress diagnostic	analysis/noise_robustness.csv	Appendix stress-test boundary only
Older corrected summary	Different analysis provenance	data/gradient_fidelity_vs_ie_corrected_summary.csv	Not used for current claims

Table S1: **Noise and gradient diagnostic provenance.** The active manuscript claims use the nonnegative-input refreshes; older corrected summaries are retained in the repository for audit history but are not mixed with the current figures.

A Supplementary Results and Diagnostics

This appendix records the supporting diagnostics, implementation details, and task protocols needed to interpret and reproduce the paper’s main claims.

Appendix structure. The appendix is organized to make the paper reproducible from mechanism to implementation. Appendix A first reports the supplementary diagnostics that support or delimit the main empirical claims. Appendix B then specifies the implemented dendritic networks, including morphology, synapse banks, shunting and additive forward equations, positive parameterization, reactivation, decoder errors, and optimizer-level gradient assignment. Appendix C collects learning-rule variants and auxiliary theory details that are useful for reproducing the local updates but are not needed for the main proof. Appendix D gives task construction and robustness protocols.

Boundary diagnostics. Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (9) is not used as a positive main result, and the pathway-vector feedback variant does not beat the best unstructured low-rank cue-routing control. These diagnostics delimit the claim rather than smoothing over failure modes.

Result Provenance and Roadmap

The main text uses only the nonnegative-input diagnostic refreshes for the gradient, noise-resilience, and path-transport claims. An older summary, `data/gradient_fidelity_vs_ie_corrected_summary.csv`, belongs to a different analysis provenance and is not used for the current main claims; in particular, its large noise-resilience additive–shunting gap should not be compared to the active nonnegative-input summaries. Tables S1–S3 make the data lineage explicit.

Diagnostic Metrics

The gradient-fidelity diagnostics compare LocalCA and backpropagation gradients at matched parameters and batches. For a parameter block p , let g_p^{local} and g_p^{bp} be the two gradients and let $w_p = \text{numel}(p) / \sum_q \text{numel}(q)$. The reported weighted cosine is

$$C_w = \sum_p w_p \frac{\langle g_p^{\text{local}}, g_p^{\text{bp}} \rangle}{\|g_p^{\text{local}}\|_2 \|g_p^{\text{bp}}\|_2 + \varepsilon},$$

and weighted scale mismatch is

$$S_w = \sum_p w_p \left| \log_{10} \frac{\|g_p^{\text{local}}\|_2 + \varepsilon}{\|g_p^{\text{bp}}\|_2 + \varepsilon} \right|.$$

Claim / diagnostic	Script anchor	Cached source	Figure / table
Exact factorization reconstructs autograd	scripts/measure_theory_diagnostics.py; scripts/generate_neurips_figures.py	analysis/theory_diag_gradient_fidelity_vs_ie_nonnegativeinput_fix_summary/theory_diag_by_condition.csv	Fig. 2A; supplementary gradient figures
Path-gain and exact-error compressibility	scripts/generate_theory_diagnostics_figures.py; scripts/measure_error_rank_diagnostics.py	analysis/error_rank_selected_20260427/error_rank_summary.csv	Fig. 3A–B; Fig. S11
Inhibitory-conductance intervention	scripts/measure_inhibition_causality.py; scripts/generate_theory_diagnostics_figures.py	analysis/inhibition_causality_selected_20260427/inhibition_causality_runs.csv	Fig. 3C; Fig. S11
Transported-error oracle	scripts/summarize_path_transport_sweep.py; scripts/generate_theory_diagnostics_figures.py	analysis/path_transport_upper_bound_nonnegativeinput_fix_5seed/path_transport_upper_bound_summary.csv	Fig. 3D–E
Rule and feedback controls	scripts/generate_neurips_figures.py	analysis/rank_bridge_nonnegativeinput_fix/noise_resilience_rank_bridge_summary.csv and rule-control summaries	Appendix Fig. S1; Table S9
Capacity-calibrated competence	scripts/summarize_neurips_claim_sweeps.py; scripts/generate_neurips_figures.py	figures/data/competence_summary_20260422.csv; Fashion-MNIST competence summaries	Fig. 4; Table S5

Table S2: **Claim provenance.** Script anchors identify the analysis entry points or figure generators used to produce each cached diagnostic.

Figure / control	Primary message	Supporting appendix material
Fig. 1	Model, factorization, and claim-type tags	Implementation equations; Table S10; Algorithm 1
Fig. 2	Exact reconstruction is precise; cosine is a weak diagnostic; scale mismatch improves	Fig. S4; Table S1; gradient metric definitions
Fig. 3	Path gains, exact-error rank, inhibition intervention, and transported-oracle chain	Fig. S11; Table S8; CIFAR and input-mode diagnostics
Fig. 4	Capacity-calibrated performance and regime dependence	Table S5; morphology map; additive-normalization and stress-test figures
Appendix Fig. S1	5F is practical, scalar broadcast matters, and more feedback bandwidth helps	Table S4; Table S9; cue-routing diagnostics

Table S3: **Main-to-supplement roadmap.** The appendix is organized around the evidence needed to interpret each main figure rather than as a development log.

480 For exact-error compressibility, we form a matrix E whose rows are examples and whose columns
481 are compartment–neuron coordinates of the exact pre-reactivation compartment error. If $E = U\Sigma V^\top$
482 and σ_i are the singular values, the rank-1 residual and participation rank are

$$\rho_1 = \frac{\sqrt{\sum_{i>1} \sigma_i^2}}{\sqrt{\sum_i \sigma_i^2}}, \quad r_{\text{part}} = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4}.$$

483 Path-gain dispersion is the coefficient of variation of α_n^{cond} over the measured sample and compart-
484 ment field. Compartment-error fidelity is the cosine between the tested broadcast field and the exact
485 compartment-error field after flattening over the same sample and compartment axes.

486 Rule and Feedback Controls

487 These controls separate three questions that are deliberately collapsed in the main competence
488 summary: which local-rule wrapper is practical, whether the somatic error source matters, and how
489 performance changes as the feedback field gains bandwidth or path information.

490 Rule-Family Ranking

491 This table keeps the 3F/4F/5F optimizer distinction explicit: the theorem-derived eligibility is
492 necessary for the story, but the reported LocalCA competence results use the empirical 5F stabilizer.

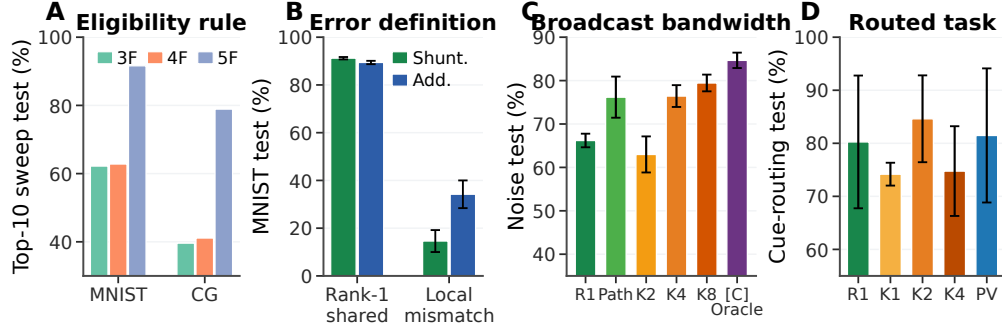


Figure S1: **Rule and feedback controls.** (A) Eligibility-rule ablation: 3F, 4F, and 5F. (B) Error-source ablation: rank-1 somatic error vs. local mismatch. (C) Feedback-bandwidth ladder on noise resilience: R1, recursive Path, K2/K4/K8, and [C] transported Oracle. (D) Cue routing: rank-1, random low-rank K1/K2/K4, and pathway-vector feedback.

Dataset	Rule	Top-10 sweep valid	Top-10 sweep test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S4: **Rule-family ranking.** Mean over the ten best validation-selected configurations from local-competence sweeps; this is not top-10 classification accuracy.

5F Stabilizer Sensitivity

Because the headline LocalCA runs use the empirical 5F stabilizer, we also tested whether performance depends sharply on the confidence clamp or the EMA rate used to estimate branch statistics. Across three-seed MNIST sensitivity checks, tightening or widening the clamp and varying the EMA rate leaves test accuracy near the default setting. This does not make 5F theorem-derived; it supports interpreting it as a bounded local reliability preconditioner rather than a brittle tuned error source.

Local Competence and Regime Dependence

Table S5 reports the capacity-calibrated headline competence numbers used in the main text. The point of this table is not to claim state-of-the-art benchmark performance, but to show the remaining LocalCA-to-backprop gap under matched dendritic capacity and rank-1 5F broadcast.

Extended Gradient Analysis and N_I Sweep Detail

Figure 2 in the main text summarizes the final-state gradient diagnostic. The supplementary view below separates static scale checks, inhibitory dose-response curves, and individual Fashion-MNIST seed behavior so that the aggregate claims are not driven by a single summary panel.

Alignment dynamics are not a zero-gradient artifact. To check whether low additive alignment could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout training (weighted norm range 9.5×10^{-4} – 9.5×10^{-3}), and the corresponding backprop gradients are also nonzero (2.3×10^{-5} – 5.7×10^{-3}). The low additive cosine therefore reflects directional and scale mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018 to 0.005), while its local/backprop norm ratio falls from about 8.8×10^2 to 0.69.

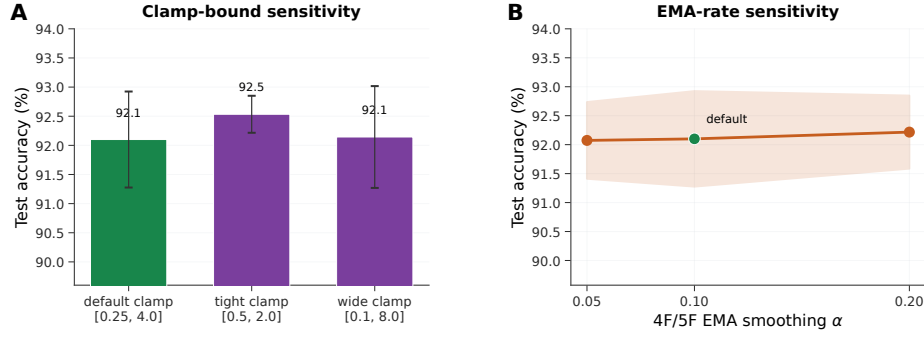


Figure S2: **5F stabilizer sensitivity.** (A) Changing the confidence-factor clamp from the default [0.25, 4.0] to tighter [0.5, 2.0] or wider [0.1, 8.0] bounds leaves MNIST test accuracy near 92%. (B) Changing the 4F/5F EMA smoothing rate over $\alpha \in \{0.05, 0.10, 0.20\}$ gives similarly stable performance. Error bars and bands are ± 1 s.d. across 3 seeds.

Dataset	BP ceiling	Shunting local	Additive local	BP gap
MNIST	0.971	0.912 ± 0.005	0.894 ± 0.007	5.9%
Fashion-MNIST	0.889	0.838 ± 0.003	0.831 ± 0.004	5.1%
Context gating	0.861	0.803 ± 0.006	0.787 ± 0.007	5.8%

Table S5: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with rank-1 broadcast and local decoder updates. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; see the HSIC auxiliary-objective appendix paragraph). Error bars on local values: ± 1 s.d. across 5 seeds.

514 Verification and Reproducibility

515 These checks compare the main seed set with held-out seeds and report the HSIC-weight sensitivity
 516 for context gating. They are included to distinguish genuine seed robustness from the known effect of
 517 removing the HSIC auxiliary term in that task.

518 Additional Stress Tests

519 These stress tests probe two ways in which low-bandwidth broadcast can become unreliable: deeper
 520 dendritic paths and noisy teaching signals. They are not separate headline benchmarks; they delimit
 521 where shunting remains useful and where additive local learning degrades.

522 FA/DFA Baseline Comparison

523 The feedback-alignment baseline is included as an adjacent local-learning reference point. DFA can
 524 be evaluated on the dendritic cores, whereas the FA condition is marked unavailable for the block-
 525 structured dendritic updates because the required random feedback matrices are not dimensionally
 526 compatible with the branch-level parameterization.

527 CIFAR-10 Results

528 Table S6 provides the exact values for the appendix CIFAR-10 figure. CIFAR-10 is used only as
 529 a harder-data mechanistic diagnostic: under flattened inputs, nonnegative conductance constraints,
 530 and a compact passive dendritic core, the matched backprop ceiling is the relevant reference rather
 531 than modern vision accuracy. In this stronger compact direct-I-stream family, shunting performs well
 532 under standard training (49.5% vs. 48.3% for additive), so the harder-data stress test does not unfairly
 533 penalize the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware
 534 soma mapping is used: additive LocalCA reaches 25.5% under rank-1 per-soma broadcast and
 535 32.4% under transported error, while shunting rises from 29.7% under rank-1 broadcast to 35.0%
 536 with random low-rank $K=4$ and 47.2% with transported error, nearly matching the 49.5% shunting
 537 backprop ceiling. Removing learned I-to-E conductance lowers the shunting ceiling by 5.7 pp and

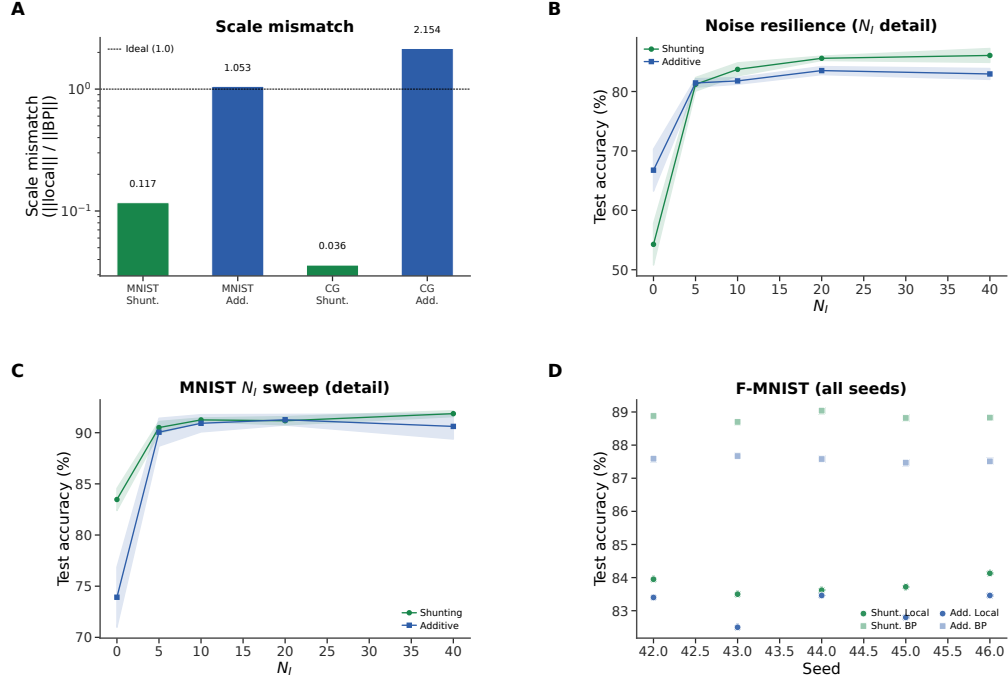


Figure S3: **Extended gradient and N_I analysis (supplementary).** (A) Scale mismatch bars from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. Shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

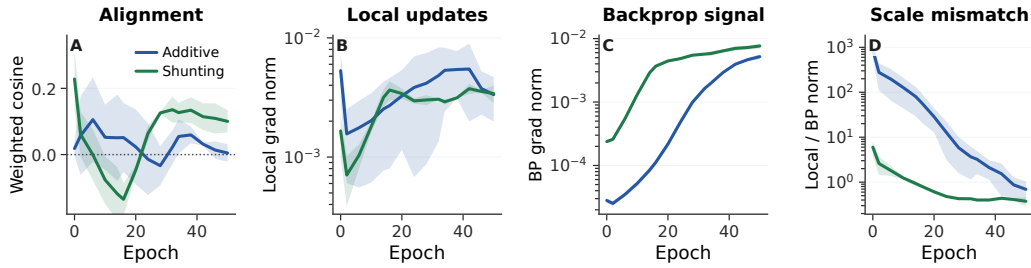


Figure S4: **Alignment dynamics with explicit gradient-norm checks.** (A) Weighted LocalCA to backprop cosine over training for the 5F rule. (B,C) Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. (D) The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

538 transported LocalCA by 8.6 pp, making this a harder-data diagnostic for inhibitory conductance as
 539 part of the useful credit geometry rather than just an added architectural detail.

540 Additive Normalization Control

541 This control separates generic additive voltage normalization from shunting conductance dynamics in
 542 a lower-capacity MNIST stress regime.

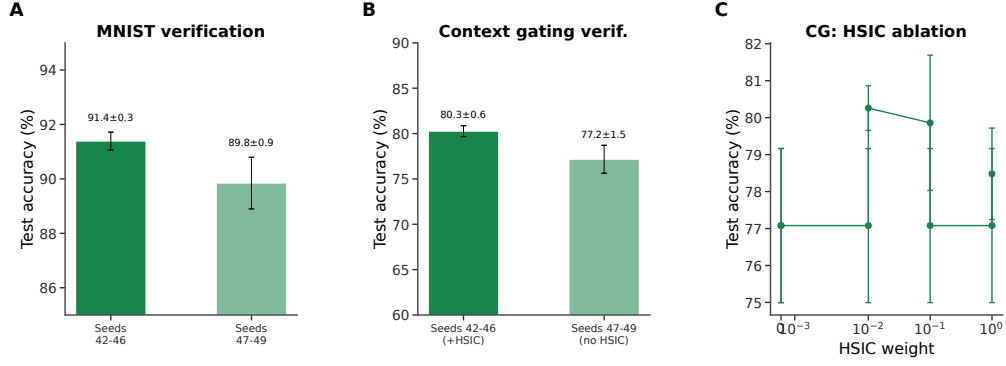


Figure S5: **Verification and reproducibility (supplementary).** (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

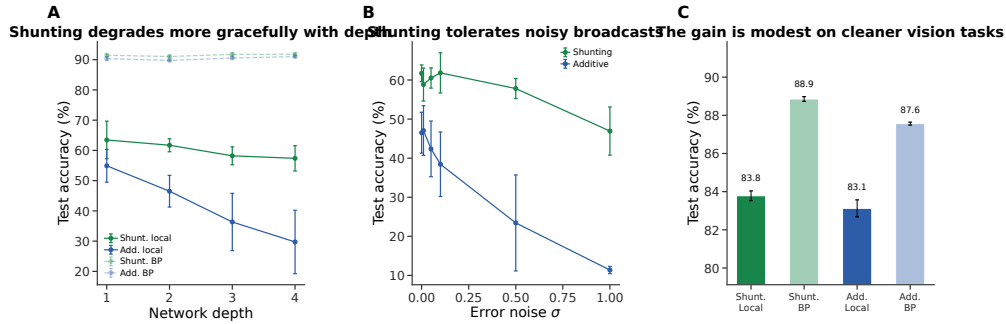


Figure S6: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

543 Morphology $\times$ Inhibition Regime Map

544 This regime map tests whether tree geometry changes the inhibitory operating point at which shunting helps. It supports the claim that morphology changes the useful inhibition regime; a fuller mechanistic
545 account would require matched theory diagnostics for every cell in the grid.
546

547 Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

548 The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative
549 excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned
550 I-to-E conductance banks directly. To test whether the same mechanism can be generated by an
551 explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe
552 that preserves the relevant dendritic depth ([3, 3] morphology). In this setting, explicit inhibitory cells
553 reach $94.2 \pm 0.2\%$ under path-transport LocalCA when their dendrites use the same update policy as
554 excitatory dendrites, $93.8 \pm 0.2\%$ when those inhibitory-cell updates are frozen, and $95.1 \pm 0.3\%$
555 under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder
556 on the same one-layer architecture: rank-1 broadcast reaches $85.2 \pm 0.7\%$, while exact effective path
557 transport reaches $95.2 \pm 0.0\%$. Thus explicit inhibitory cells can generate useful path-gain structure
558 once the architecture isolates the dendritic-path question.

559 **Post-training inhibition intervention and exact-error rank.** We next tested whether learned
560 inhibitory conductance carries task-specific structure rather than only changing global scale. We

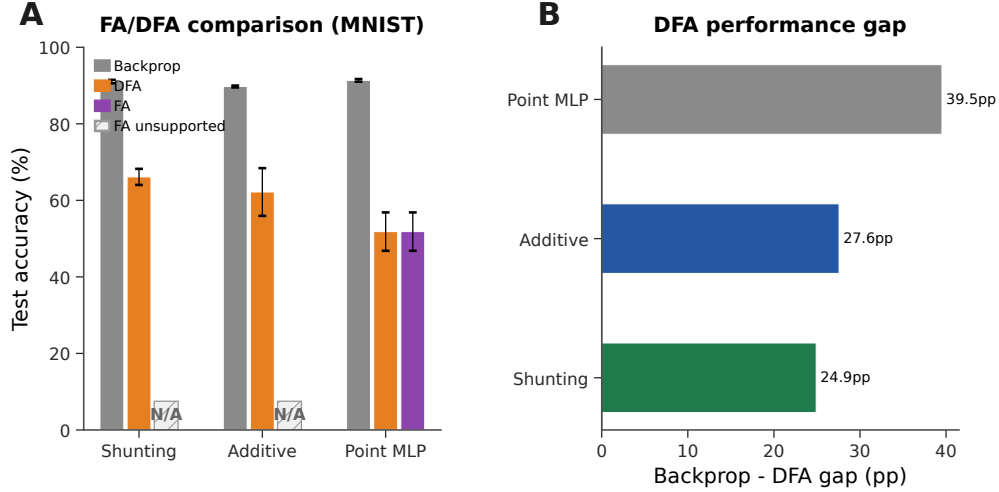


Figure S7: **Feedback alignment baselines (MNIST)**. (A) Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

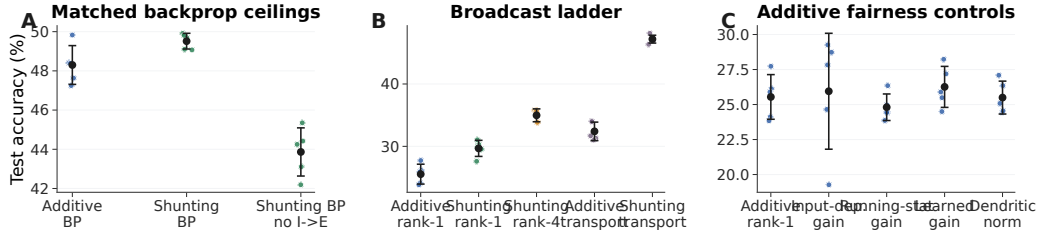


Figure S8: **CIFAR-10 control ladder in the compact direct-I-stream family**. (A) Matched backprop ceilings. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting ceiling. (B) Broadcast ladder. Rank-1 broadcast remains weak on this harder dataset, rank-4 improves partially, and exact effective transported error nearly reaches the shunting backprop ceiling. (C) Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive rank-1 broadcast. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

561 replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conduc-
 562 tance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch
 563 batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST
 564 inhibition at $N_I=5$ gives $90.4 \pm 1.1\%$ accuracy, while the four interventions drop to 19.8–26.4%.
 565 On noise resilience at $N_I=10$, learned inhibition gives $80.5 \pm 2.4\%$, while the interventions drop
 566 to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance
 567 load. We also computed the SVD of exact compartment-error matrices across samples and compart-
 568 ments. At matched MNIST $N_I=5$, shunting has lower rank-1 residual than additive (0.65 ± 0.18
 569 vs. 0.83 ± 0.07) and lower participation rank (3.2 ± 1.2 vs. 8.5 ± 3.3), supporting the path-gain
 570 compressibility interpretation.

571 We also computed the direct log-gain covariance margin in Eq. (9) on the selected morphology
 572 diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly
 573 negative in the inhibited shunting selections (-2.5×10^{-3} , -5.0×10^{-4} , and -2.8×10^{-4} at
 574 $N_I=5, 20, 40$). We therefore do not use this covariance condition as a positive main result. Its

Condition	Core	Training / broadcast	I-to-E	Test acc.
standard	additive	backprop	yes	48.3 ± 1.0
standard	shunting	backprop	yes	49.5 ± 0.4
standard, no learned inhibition	shunting	backprop	no	43.9 ± 1.2
rank-1	additive	5F per-soma	yes	25.5 ± 1.6
rank-1, input-dependent gain	additive	5F per-soma	yes	25.9 ± 4.1
rank-1, running-stat gain	additive	5F per-soma	yes	24.8 ± 0.9
rank-1, learned gain	additive	5F per-soma	yes	26.3 ± 1.5
rank-1, dendritic normalization	additive	5F per-soma	yes	25.5 ± 1.2
rank-1	shunting	5F per-soma	yes	29.7 ± 1.3
rank-1, no learned inhibition	shunting	5F per-soma	no	22.3 ± 0.7
transported error	additive	5F path transport	yes	32.4 ± 1.5
rank-4	shunting	5F low-rank	yes	35.0 ± 1.0
transported error	shunting	5F path transport	yes	47.2 ± 0.6
transported error, no learned inhibition	shunting	5F path transport	no	38.6 ± 1.0

Table S6: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in $[0, 1]$, nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backprop ceiling, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive rank-1 fairness controls do not explain the transported-shunting result.

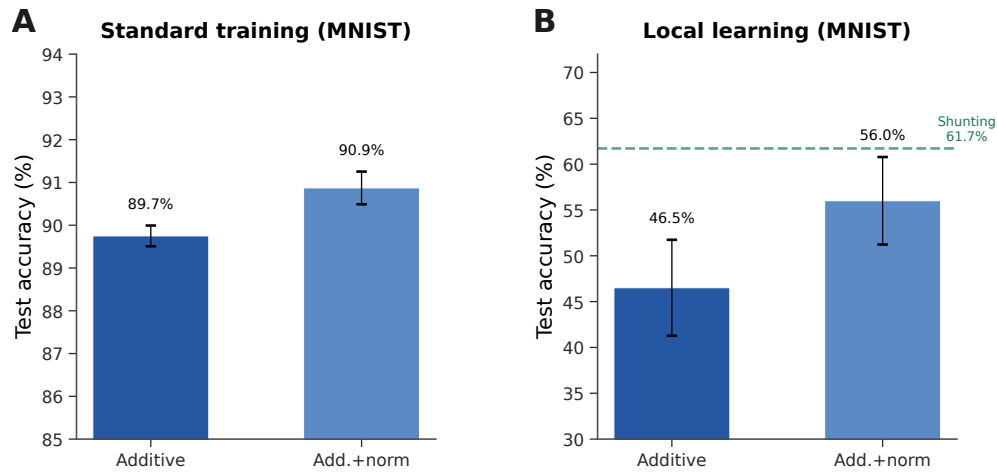


Figure S9: **Additive + normalization control (MNIST stress-regime diagnostic).** (A) Standard backprop: normalization provides a small boost (89.7% → 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). These lower local accuracies come from a separate stress-regime diagnostic, not from the capacity-calibrated competence setting in Table S5.

role is to clarify when shunting should compress gains; the empirical claim rests on the measured comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

Experimental Predictions

The model is not a detailed cell-type circuit model, but its path-gain identity gives falsifiable dendritic predictions. These predictions are naturally aligned with branch-level dendritic computation and compartment-targeted inhibition in pyramidal neurons [5–7, 12].

Cue-Routing Feedback-Rank Diagnostic

Cue routing tests the regime where one shared teaching signal is expected to fail: context determines which noisy cue is reliable, so different pathways should receive different credit. Figure S12 shows

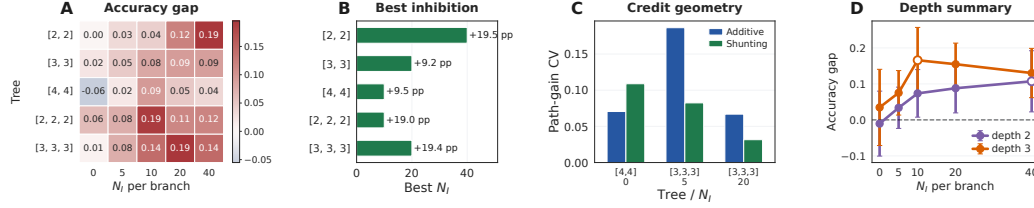


Figure S10: **Morphology shifts the useful inhibitory regime.** (A) Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. (B) Best inhibitory count for each morphology. (C) Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. (D) Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

Inhibitory input mode	Training / broadcast	Test accuracy
Direct inhibitory stream	LocalCA, per-soma rank-1	0.852 ± 0.007
Direct inhibitory stream	LocalCA, exact effective path transport	0.952 ± 0.000
Explicit I cells	LocalCA, path transport, train I-cell dendrites	0.942 ± 0.002
Explicit I cells	LocalCA, path transport, freeze I-cell dendrites	0.938 ± 0.002
Explicit I cells	standard backprop	0.951 ± 0.003

Table S7: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with [3, 3] morphology and nonnegative inputs. The direct-stream rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with [3, 3] morphology. Exact effective path transport closes the direct-stream rank-1 gap, and explicit inhibitory cells nearly match both direct-stream path transport and the explicit-I backprop ceiling.

that the benchmark is learnable and that moving beyond rank-1 feedback helps, but the path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback method.

Random Low-Rank Broadcast Channels

These controls test whether failures of rank-1 broadcast reflect insufficient feedback bandwidth rather than a failure of local eligibility itself.

B Dendritic Network Implementation

This section gives the implementation-level specification needed to reproduce the dendritic architectures and LocalCA updates. It separates the biological modeling assumptions from software choices such as positive raw-parameter transforms, decoder-error coordinates, and optimizer gradient assignment.

Implementation Overview

Each feedforward dendritic layer is a population of dendritic neurons. A neuron contains a rooted tree of compartments; each compartment has an excitatory synapse bank, optionally an inhibitory synapse bank, and dendritic coupling conductances from its children. The main experiments use branch-level input-driven inhibition: a nonnegative transfer stage produces excitatory and inhibitory input streams from the external input, and the inhibitory stream drives learned I-to-E conductance banks on excitatory dendritic branches. The explicit-I probe in Table S7 instead instantiates a separate feedforward inhibitory population, whose output projects inhibitory conductance onto excitatory dendrites.

The additive and shunting cores are architecture-matched: they use the same tree, synapse counts, nonnegative transfer stage, positive conductance parameterization, optimizer protocol, and seed sets.

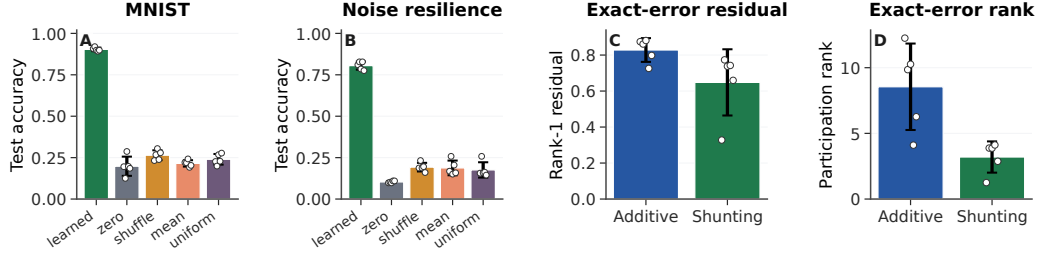


Figure S11: **Post-training inhibition intervention and exact-error compressibility.** (A,B) Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field carries sample-specific task structure. (C,D) SVD diagnostics of the exact compartment-error matrix on matched MNIST $N_I=5$ runs. Shunting has a smaller rank-1 residual and lower participation rank than additive, so the exact error field is more compressible. Error bars are s.d. across 5 seeds.

Model implication	Biological prediction
Path-local inhibitory conductance changes α_n^{cond} only for descendants whose credit path crosses the inhibited compartment.	Branch-local inhibitory perturbation should distort plasticity or teaching-signal efficacy for synapses below the perturbed compartment more than for sister branches.
The log derivative $\partial \log \alpha_n^{\text{cond}} / \partial G_k^I = -R_k^{\text{tot}}$ predicts stronger credit gating in high-resistance compartments.	The same inhibitory conductance change should have larger effects on low-conductance/high-resistance branches than on already high-conductance branches.
Rank-1 broadcast succeeds when the exact compartment-error matrix is low-rank and fails when branch identity matters.	Tasks requiring route- or context-specific branch credit should need richer dendritic teaching signals than a single scalar modulatory broadcast.

Table S8: **Experimental predictions from the path-gain mechanism.** These are qualitative biological predictions, not claims tested directly in this paper.

They differ only in how a branch combines excitatory, inhibitory, and dendritic inputs. Decoder weights are unconstrained readout parameters and are not interpreted as conductances.

Morphology and Synapse Banks

The morphology notation $[b_1, b_2, \dots, b_D]$ specifies a rooted dendritic tree. The factor b_d is the fan-in from dendritic stage d to the next more proximal stage. Thus $[3, 3]$ gives 3 proximal branches per soma and 3 distal branches per proximal branch, for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic branches rather than directly on the soma. Unless a control explicitly changes this, each branch has N_E excitatory synapses and N_I inhibitory synapses. The branch input is sparse through a learned top- k or banked synapse selection module; inactive synapses are masked out in the local update unless the corresponding control enables inactive-weight updates.

For a branch n , write $x_{nj}^E, x_{nj}^I \geq 0$ for the selected excitatory and inhibitory presynaptic drives and a_c for the post-reactivation activity of child compartment c . The branch-level excitatory, inhibitory, and dendritic conductance currents are

$$U_n^E = \sum_{j=1}^{N_E} g_{nj}^E x_{nj}^E, \quad U_n^I = \sum_{j=1}^{N_I} g_{nj}^I x_{nj}^I, \quad D_n = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}} a_c.$$

The corresponding dendritic coupling load is

$$G_n^{\text{den}} = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}}.$$

For leaf branches, $D_n = G_n^{\text{den}} = 0$. For direct input-driven inhibition, x^I is produced by the inhibitory transfer stream. For explicit-I controls, x^I is the activity of the inhibitory-cell population.

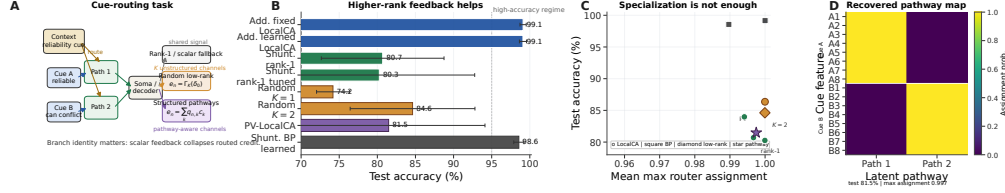


Figure S12: **Structured feedback when rank-1 broadcast fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of rank-1, random low-rank, and pathway-structured feedback. Moving beyond rank-1 helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

Setting	Broadcast	Test accuracy
<i>Noise resilience, shunting, rank bridge</i>		
per-soma	rank-1 per-soma	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K=1$	0.453 ± 0.126
low-rank	$K=2$	0.630 ± 0.042
low-rank	$K=4$	0.764 ± 0.025
low-rank	$K=8$	0.795 ± 0.019
path transport	exact effective path transport	0.847 ± 0.018
<i>Cue routing, learned shunting router, rank/structure sweep</i>		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	$K=1$	0.742 ± 0.022
low-rank	$K=2$	0.846 ± 0.082
low-rank	$K=4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S9: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both recursive path propagation and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact effective path transport, with low-rank continuing to improve up to $K=8$. On cue routing, the learned-router sweep shows that moving beyond rank-1 helps, but the best unstructured low-rank setting ($K=2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

623 Shunting and Additive Forward Equations

624 In the shunting core, excitatory reversal is 1, leak and inhibitory reversal are 0, and leak conductance
625 is fixed to 1. The implemented branch voltage is therefore

$$V_n^{\text{shunt}} = \frac{U_n^E + D_n}{1 + U_n^E + U_n^I + G_n^{\text{den}} + \varepsilon}.$$

626 The small numerical ε is used only for floating-point safety; analytically the leak conductance keeps
627 the denominator positive. The total conductance and input resistance recorded for learning are

$$g_n^{\text{tot}} = 1 + U_n^E + U_n^I + G_n^{\text{den}}, \quad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}.$$

628 In the additive control, inhibition is subtractive rather than shunting. With the same nonnegative
629 conductance variables and nonnegative inputs, the implemented comparator is

$$V_n^{\text{add}} = U_n^E + D_n - U_n^I,$$

630 with optional additive-normalization controls reported separately in Fig. S9. Thus additive inhibition
631 changes the signed voltage contribution and local additive eligibility, but it does not enter the
632 denominator and does not gate conductance-stage path gain through R_n^{tot} .

Paper object	Code anchor	Role
Shunting branch voltage	<code>DendriticBranchLayer.forward</code> with <code>use_shunting=True</code>	Implements numerator/denominator conductance integration and stores total conductance for diagnostics.
Additive control voltage	<code>DendriticBranchLayer.forward</code> with <code>use_shunting=False</code> ; <code>normalize_additive_voltage</code> for the normalization control	Uses the same branch tree and synapse banks, but inhibitory input is subtractive rather than a denominator load.
Broadcast-mode switch	<code>LocalCreditAssignment._apply_local_rule_gradients</code> ; <code>LocalRuleConfig.error_broadcast_mode</code>	Chooses scalar/per-soma, local-mismatch, low-rank, path-transport, and pathway-vector feedback fields.
Low-rank feedback	<code>LocalCreditAssignment._compute_low_rank_broadcast</code> ; <code>broadcast_rank</code>	Fixed random low-rank broadcast controls used in the bandwidth ladder.
Path-transport oracle	<code>LocalCreditAssignment._precompute_path_transport_errors</code>	[C] transported-error upper bound using the effective tree recursion.
Path-propagation proxy	<code>LocalCreditAssignment._precompute_path_propagation_factors</code>	Approximate morphology-aware attenuation control, distinct from exact path transport.
Shunting eligibility	<code>LocalCreditAssignment._apply_local_rule_gradients</code> conductance branch	Assigns $xR^{\text{tot}}(E - V)e$ gradients, then multiplies by the positive-transform derivative.
Additive eligibility	<code>LocalCreditAssignment._apply_local_rule_gradients</code> additive branch	Assigns the matched signed additive derivative, with optional additive-gain controls.
4F / 5F reliability gates	<code>_compute_layer_rho</code> , <code>_compute_layer_phi*</code> , <code>clamp_phi</code> , <code>FiveFactorConfig</code>	Slow reliability preconditioners used by 4F/5F, not additional fast task-error channels.
Experiment knobs	<code>LocalRuleConfig</code> ; YAML fields <code>rule_variant</code> , <code>error_broadcast_mode</code> , <code>broadcast_rank</code> , <code>five_factor</code>	Reproduces the rule family, feedback bandwidth, and 5F stabilizer settings.

Table S10: **Equation-to-code map**. Code anchors are symbolic names rather than line numbers so that minor refactors do not invalidate the reproducibility map.

633 Positive Parameterization and Raw-Parameter Gradients

634 Trainable synaptic and dendritic conductances are stored as unconstrained raw parameters θ and
635 transformed by a positive map $g = \psi(\theta)$, with `softplus` used in the main reported runs:

$$\psi(\theta) = \log(1 + \exp \theta), \quad \psi'(\theta) = \sigma(\theta).$$

636 Equations in the main text give conductance-space gradients $\partial L / \partial g$. Before assigning gradients to
637 the optimizer variables, the implementation applies the chain rule

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial g} \psi'(\theta).$$

638 The exact-gradient reconstruction diagnostic includes this parameterization factor, as well as top- k
639 activity masks and dendritic block-linear parameterization. This is why the reconstruction comparison
640 is against raw autograd gradients rather than only against abstract conductance variables.

641 Post-Voltage Reactivation

642 The conductance stage produces a pre-reactivation voltage V_n , after which each branch may transmit a
643 monotone post-voltage activity $a_n = f_n(V_n)$. The main sweeps use the learnable bounded transform

$$f_n(V) = \frac{\tanh(m_n(V - b_n)) + 1}{2}, \quad m_n = \exp(\ell_{m,n}),$$

644 with derivative

$$f'_n(V) = \frac{m_n}{2} [1 - \tanh^2(m_n(V - b_n))].$$

645 When reactivation is enabled, exact path transport uses the effective gain $\tilde{\alpha}_n$: parent-to-child transport
646 includes the parent activation derivative, and the local update converts activation-space error into
647 pre-reactivation voltage error by multiplying by the child derivative. With identity transfer, this
648 reduces to conductance-stage transport through α_n^{cond} .

Quantity	Symbol	Convention
Voltage	V	Conductance-stage shunting voltage in $[0, 1]$ under nonnegative drive; additive controls may be signed
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonnegative via $g = \text{softplus}(\theta)$; local gradients are mapped to raw θ by the transform derivative
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S11: Units and normalization.

Decoder and Error Coordinates

Let h denote the final soma/core activity passed to the decoder and let $\delta^y = \partial L / \partial \hat{y}$. For a linear decoder $\hat{y} = W_{\text{dec}} h$, the soma/core error is

$$\delta_0 = W_{\text{dec}}^\top \delta^y.$$

For a nonlinear decoder, used in the CIFAR-10 compact control ladder, the implementation uses the decoder-input Jacobian product

$$\delta_0 = J_{\text{dec}}(h)^\top \delta^y.$$

All LocalCA broadcast modes are defined in this soma/core error coordinate system. Decoder-update modes are: **backprop**, which uses autograd for W_{dec} ; **local**, which applies $\Delta W_{\text{dec}} \propto \langle \delta_0 h^\top \rangle_B$; and **frozen**, which holds the decoder fixed.

Biological Plausibility Assumptions

The model makes the following explicit assumptions. First, it uses the steady-state solution of the passive cable equation rather than temporal membrane dynamics; all path gains and gradient diagnostics refer to conductance-stage voltages at steady state. Second, each dendritic cell is a rooted tree, so each compartment has a unique path to the soma and the closed-form path gains α_n^{cond} and $\tilde{\alpha}_n$ are well-defined. Third, synaptic and dendritic conductance parameters are nonnegative by construction. Fourth, the main sweeps enforce nonnegative first-layer presynaptic drive through nonnegative inputs or a ReLU transfer stage. Fifth, post-voltage reactivation is handled through local $f'(V)$ factors along the path. Sixth, every synapse receives only a scalar or low-rank non-local broadcast; presynaptic activity, voltage, reversal potential, input resistance, covariance modulators, and confidence factors are compartment-local or slowly estimated local quantities.

Units and Parameterization

Table S11 summarizes the numerical conventions used throughout the implementation and diagnostics, including which quantities are constrained conductances and which are unconstrained readout parameters.

Representative Manuscript Architectures

Hyperparameters

Table S13 collects the core training hyperparameters for each main experiment. Learning rates are applied through parameter groups (top- k , dendritic block-linear, reactivation, decoder) with the values given in the representative config files; where a single LR is reported, every group uses that value.

Effective Broadcast Dimensionality in Main Architectures

Compute Resources

All experiments were run on an institutional GPU cluster. Reproduction runs use one NVIDIA A100 or H100-class GPU (40–80GB memory) per seed and do not require distributed training. Table S15 gives approximate single-GPU runtime ranges for the reported experiment families; unreported exploratory runs used additional cluster time and are not counted in the reproduction estimate.

683 Code, Data, and Asset Availability

684 MNIST, Fashion-MNIST, and CIFAR-10 are public datasets or public benchmark assets cited in the
 685 paper; we use them under their standard published access conditions and cite their original sources
 686 in the bibliography. To preserve double-blind review, the submission provides code through an
 687 anonymized supplementary ZIP rather than a public repository URL. The ZIP contains the core
 688 model and training code, selected diagnostic utilities, representative configuration files, tests, and
 689 reproduction instructions for the main experimental families; code and configurations will be de-
 690 anonymized after review. The paper does not release a new dataset or a stand-alone pretrained model
 691 asset. Dataset access pages are the original MNIST site / UCI entry (<https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits>), the Fashion-MNIST reposi-
 692 tory (<https://github.com/zalandoresearch/fashion-mnist>), and the CIFAR page (<https://www.cs.toronto.edu/~kriz/cifar.html>).

695 LocalCA Broadcast and Gradient Assignment

696 LocalCA first maps the output loss derivative to the soma/core error δ_0 and then constructs a branch
 697 error field. For the rank-1 setting used in the main competence results, the scalar broadcast is
 698 computed per example:

$$\bar{\delta}_b = \frac{1}{d_0} \sum_{c=1}^{d_0} \delta_{0,bc}, \quad e_{n,b}^a = \bar{\delta}_b \mathbf{1}_n,$$

699 where d_0 is the dimension of the soma/core error. If the layer width matches the error dimension, the
 700 per-soma mode uses $e_n^a = \delta_0$ instead. The low-rank controls use fixed random projection and mixing
 701 matrices P_K and Q_n ,

$$c_b = P_K \delta_{0,b} \in \mathbb{R}^K, \quad e_{n,b}^a = Q_n c_b.$$

702 The pathway-vector controls use router-inferred pathway roles to gate and transport a low-rank
 703 role vector. The transported oracle uses the effective tree recursion. If p is the parent of child c ,
 704 activation-space error is propagated by

$$e_c^a = e_p^a f_p'(V_p) R_p^{\text{tot}} g_{c \rightarrow p}^{\text{den}},$$

705 and the local pre-reactivation voltage error used in conductance eligibility is

$$e_n^V = \begin{cases} e_n^a f_n'(V_n), & \text{post-voltage reactivation enabled,} \\ e_n^a, & \text{identity transfer.} \end{cases}$$

706 This operational definition is the implementation counterpart of the effective path gain $\tilde{\alpha}_n$ in Theo-
 707 rem 1.

708 For shunting branches, the conductance-space LocalCA gradients assigned before raw-parameter
 709 transformation are

$$\hat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E R_n^{\text{tot}} (1 - V_n) \rangle_B, \quad \hat{\nabla}_{g_{nj}^I} L = \langle e_n^V x_{nj}^I R_n^{\text{tot}} (0 - V_n) \rangle_B,$$

710 and for a dendritic coupling from child c to parent n ,

$$\hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V R_n^{\text{tot}} (a_c - V_n) \rangle_B.$$

711 For additive controls, the matching local derivatives are

$$\hat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E \rangle_B, \quad \hat{\nabla}_{g_{nj}^I} L = -\langle e_n^V x_{nj}^I \rangle_B, \quad \hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V a_c \rangle_B.$$

712 The optimizer applies the usual descent step; equivalently, the sign can be absorbed into the definition
 713 of the broadcast error. After these conductance-space gradients are formed, they are multiplied by
 714 the positive-transform derivative $\psi'(\theta)$ and by any active synapse masks before assignment to raw
 715 parameters.

716 4F and 5F Local Modulators

717 The 3F rule uses only the local eligibility and broadcast error above. The 4F and 5F variants multiply
 718 the conductance-space gradient by local reliability factors before raw-parameter transformation. The
 719 4F factor is an online branch-soma covariance proxy,

$$r_n^{4F} = \frac{\text{Cov}(\bar{V}_n, \bar{V}_0)}{\sqrt{\text{Var}(\bar{V}_n)\text{Var}(\bar{V}_0) + \varepsilon}}.$$

720 The 5F factor adds a confidence term based on how much branch voltage variance remains predictable
 721 from parent or soma-level activity:

$$\phi_n = \text{clip}_{[0.25, 4.0]} \left(\frac{\text{Var}(V_n)}{\sigma_{\text{res},n}^2 + \varepsilon} \right),$$

722 where $\sigma_{\text{res},n}^2$ is estimated online by the configured residual-variance estimator. In the main 5F runs,
 723 the multiplier is $r_n^{4F} \phi_n$. The 5F sensitivity diagnostic in Fig. S2 varies the clamp and EMA rate to
 724 verify that the reported MNIST performance is not a single brittle clamp setting.

725 Algorithm

726 Algorithm 1 gives the update order for the main 5F rank-1 LocalCA condition and shows where
 727 higher-bandwidth feedback controls enter.

Algorithm 1 Main 5F Rank-1 LocalCA Update

- 1: **Input:** Dendritic model, batch (x, y) , learning rate η
 - 2: Forward pass; loss L , output error δ^y
 - 3: Somatic/core error $\delta_0 = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$ (or $W_{\text{dec}}^\top \delta^y$ for a linear decoder)
 - 4: **for** each layer n (reverse) **do**
 - 5: Rank-1 broadcast $e_n = \text{mean}(\delta_0) \mathbf{1}_n$
 - 6: Update EMA estimates for r_n^{4F} and ϕ_n from batch voltages
 - 7: Convert to pre-reactivation voltage error e_n^V using the local activation derivative when needed
 - 8: $\Delta g_j^{\text{syn}} \leftarrow \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n^V \rangle_B$
 - 9: $\Delta g_{c \rightarrow n}^{\text{den}} \leftarrow \eta r_n^{4F} \phi_n \langle R_n^{\text{tot}} (a_c - V_n) e_n^V \rangle_B$
 - 10: Map conductance gradients to raw parameters by multiplying by $\partial g / \partial \theta$
 - 11: **end for**
 - 12: Optional controls replace line 5 with neuron-wise, rank- K , path-structured, or transported-oracle broadcasts.
 - 13: Clip gradients; optimizer step
-

728 C Theoretical Details

729 This section collects theory-adjacent material that supports implementation and interpretation but is
 730 not part of the main proof. The central derivation remains Theorem 1, Corollary 1, and Prop. 2.

731 Random-Broadcast Alignment Intuition

732 Feedback-alignment arguments suggest that random or low-rank feedback can be useful when the
 733 induced local update remains positively correlated with the exact gradient [13]. In this model, that
 734 correlation depends on how local eligibility factors co-vary with the conductance-stage path gain in
 735 Eq. (4). We treat this as intuition only. The central theoretical object in the paper is still the exact
 736 pre-reactivation compartment error $\partial L / \partial V_n$; under identity upward transfer this reduces to $\alpha_n^{\text{cond}} \delta_0$,
 737 while post-voltage activation uses $\tilde{\alpha}_n \delta_0$. The empirical question is how faithfully different broadcast
 738 fields approximate that quantity.

739 Morphology-Aware Extensions

740 The following variants are implemented controls or architectural extensions used to test whether
 741 coarse morphology-aware feedback can approximate exact path transport. They are not required

for the main 5F rank-1 results, but they define the path-propagation, depth, normalization, and pathway-vector controls reported in the appendix figures and tables.

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. (4) without computing the exact sample-specific path gain.

Depth modulation. Per-branch scaling $\kappa_j = \kappa_{\text{base}}/(d_j + d_0)$, mirroring cable attenuation and testing whether a simple depth prior can stabilize distal updates.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}}/(\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [25] and used as an additive-control normalization comparison.

Pathway-vector feedback. For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

HSIC Auxiliary Objectives

Following [23], we use kernel-based HSIC objectives:

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (r_n^{4F} , ϕ_n) use Welford’s algorithm [24].

D Task Construction and Robustness Protocols

This section specifies how the benchmark and synthetic tasks are constructed, including which tasks preserve the nonnegative-input regime assumed by the conductance model. It also records the stress-test protocols used for depth and broadcast-noise robustness.

General Data Handling

MNIST, Fashion-MNIST, and CIFAR-10 are used through standard public dataset loaders. Images are flattened before entering the dendritic core, scaled to $[0, 1]$, and passed through the configured nonnegative transfer stage in the main experiments. MNIST-derived synthetic tasks inherit the flattened 28×28 input geometry. Random task components, such as corruption projections and context masks, are generated from fixed seeds and reused across the corresponding train, validation, and test splits. Reported validation-best results use validation accuracy for checkpoint selection and test accuracy for final reporting.

The paper does not use data augmentation as a performance device. CIFAR-10 is intentionally a flattened harder-data diagnostic under the same conductance constraints rather than a competitive vision benchmark.

Supervised Benchmark Tasks

The supervised tasks below provide the capacity-calibrated classification setting, the harder-data diagnostic, and the structured nonnegative synthetic tasks used to probe broadcast limitations.

MNIST and Fashion-MNIST. The standard classification tasks use ten output classes and non-negative pixel inputs. They serve as capacity-calibrated tests of whether rank-1 5F LocalCA can approach a matched backpropagation reference on clean supervised data. Fashion-MNIST is included as a modest distribution shift with the same input and output dimensionality as MNIST.

783 **CIFAR-10 harder-data diagnostic.** CIFAR-10 inputs are flattened RGB images in $[0, 1]$. The
 784 compact CIFAR family uses a smaller dendritic core with a nonlinear decoder and decoder-aware
 785 LocalCA error mapping. We use this setting to test whether the broadcast-fidelity ladder persists on a
 786 harder data distribution: rank-1 broadcast is weak, higher-rank feedback helps, and exact effective
 787 path transport approaches the matched shunting backprop ceiling.

788 **Context gating.** Inputs are contextual figure-ground MNIST examples used for 10-way digit
 789 classification. The right half of the image carries the original MNIST digit, while the left half is
 790 replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests
 791 whether credit assignment can exploit a simple context-dependent spatial structure without leaving
 792 the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01)
 793 is used in the main LocalCA competence runs to decorrelate layer representations, adding roughly
 794 3 pp under local learning in the ablation.

795 **Noise resilience.** Inputs are flattened MNIST images corrupted by structured channel noise: a fixed
 796 50-dimensional latent Gaussian perturbation is projected into input space and added to each example,
 797 with $\sigma_{\text{task}} = 1.5$, then clipped to $[0, 1]$. The projection is fixed across training, so the network must
 798 filter correlated interference to recover the underlying class signal. Credit assignment must remain
 799 coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

800 **Cue integration.** Two noisy cue streams (A and B) are presented simultaneously, each carrying
 801 partial information about one of 2 classes. A binary one-hot context signal indicates which cue
 802 is reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0, 1]$ after noise is
 803 added. The fixed-pathway variant duplicates the context into both cue branches; the learned-routing
 804 variant requires the router to discover the cue separation from data. This task is used in the cue-
 805 routing feedback-rank diagnostic to test structured pathway-vector broadcast while staying inside the
 806 nonnegative-input regime.

807 **Depth Scaling and Noise Robustness**

808 These robustness protocols are used only as stress diagnostics. They keep the same broad model
 809 family while changing dendritic depth or corrupting the broadcast signal.

810 **Depth scaling.** Varying dendritic depth from 1–4 layers (branch factors $[9]$ to $[3, 3, 3, 3]$) tests
 811 whether local credit deteriorates with longer paths. Shunting local degrades from 63.5% to 57.4%;
 812 additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth
 813 ($+8.5 \rightarrow +27.7$ pp). Backprop ceilings remain stable at about 90–92%. See Fig. S6A.

814 **Noise robustness.** Gaussian noise $\mathcal{N}(0, \sigma^2)$ is added to the broadcast error to test whether learning
 815 depends on a finely tuned scalar magnitude. Shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive
 816 drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals carry useful learning
 817 information beyond the broadcast magnitude. See Fig. S6B.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	[128, 128]	[3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .
MNIST / context-gating local competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Best local 5F per-soma runs in Table S5; reactivation <code>param_tanh</code> .
Fashion-MNIST competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep; reactivation <code>param_tanh</code> .
Cue-routing PV-LocalCA	router	[64]	[2]	$N_E=10, N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .
CIFAR-10 compact direct-I-stream control ladder	id.+ReLU	[20] E, no I cells	[3, 3, 3, 3]	$N_E=25, N_I=25$ direct I-to-E, with matched no-I controls	5 per condition	Harder-data extension with input-driven inhibitory conductance, decoder [32, 16], decoder-aware soma mapping, and additive fairness controls.
Morphology $\times$ inhibition appendix map	id.+ReLU	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	3	Supportive 3-seed sweep; reactivation <code>param_tanh</code> .

Table S12: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks without direct somatic synapses, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

Setting	Opt.	LR	Batch	Epochs	W-dec.	Notes
MNIST / Fashion-MNIST / context gating (5F per-soma LocalCA)	Adam	$10^{-3} / 5 \cdot 10^{-4}$ (block / react.)	256	100	0	fixed-epoch, no early stopping; <code>param_tanh</code> reactivation
MNIST / Fashion-MNIST / context gating (BP ceiling)	Adam	$10^{-3} / 5 \cdot 10^{-4}$	256	100	0	matched to LocalCA setup
Gradient fidelity / path-gain / noise resilience	Adam	10^{-3}	256	50	0	5 seeds, hooks enabled for diagnostic capture
Morphology $\times$ inhibition regime map (supportive)	Adam	10^{-3}	256	50	0	3 seeds
Cue routing	Adam	10^{-3}	256	90	0	early stopping with patience 30
CIFAR-10 compact direct-I-stream depth-4 LocalCA	Adam	$10^{-3} / 10^{-4}$ (block / react.)	256	400	0 / 0.01	grad-clip 5.0, early stop patience 50, nonlinear decoder
CIFAR-10 compact direct-I-stream depth-4 BP	Adam	10^{-3}	256	200	0.01	early stop patience 40, matched ceiling

Table S13: **Training hyperparameters by experiment.** All runs use Adam with default $\beta_1=0.9$, $\beta_2=0.999$, $\varepsilon=10^{-8}$. Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Representative configs are provided in the anonymized supplementary ZIP.

Setting	Soma/core dim.	Output classes	Main feedback field
Gradient fidelity / path-gain / noise resilience	128, 128	10	Explicit scalar compression of soma/core error at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Explicit scalar compression of soma/core error
Context gating competence	128	10	Explicit scalar compression of soma/core error
Cue routing	64	2	Explicit scalar compression by default; higher-rank tests use low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	10	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S14: **Effective dimensionality of the main broadcast in the reported architectures.** Output loss gradients are first mapped into decoder-input/soma coordinates. Main rank-1 experiments then deliberately compress that soma/core error to one scalar per example and broadcast it to all compartments; vector-valued per-soma, low-rank, pathway-vector, and transported-error conditions are labeled as higher-bandwidth controls.

Experiment family	Runs	Time / run	Approx. GPU-h
Competence and verification	~25	5–20 min	4–9
Gradient, path-gain, inhibition, and oracle diagnostics	~80	10–30 min	15–40
Morphology, stress, rule, and feedback controls	~120	5–35 min	20–70
CIFAR-10 control ladder	70	0.8–2.5 h	60–175
Figure generation and CPU-side summaries	–	< 5 h total	< 5

Table S15: **Approximate compute for reported results.** Ranges reflect single-seed wall-clock variation across A100/H100-class GPUs, data-loading overhead, and early stopping.

Asset	Use in this paper	Access / license or terms note
MNIST	Digit classification and derived nonnegative synthetic tasks	Public benchmark from the original MNIST site / UCI entry; UCI lists DOI 10.24432/C53K8Q and asks users to follow the original acknowledgement policy; no raw-data redistribution.
Fashion-MNIST	Apparel classification stress test	Public Zalando Research benchmark; repository is MIT licensed; no raw-data redistribution beyond standard dataset loaders.
CIFAR-10	Flattened harder-data diagnostic	Public CIFAR dataset site asks users to cite Krizhevsky’s technical report; no raw-data redistribution.
Synthetic tasks	Context gating, noise resilience, cue integration	Generated procedurally from public benchmarks or random seeds described in Appendix D; no new third-party asset is introduced.

Table S16: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state the paper’s main mechanistic claim, the scope of the conductance-based setting, and the regime-dependent nature of the empirical results; see Sec. 1 and Sec. 5.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The discussion explicitly narrows the scope of the claims and discusses limitations involving benchmark strength, broadcast assumptions, depth scaling, morphology, activation shape, and feedback rank; see Sec. 5.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The voltage model, assumptions, theorem statements, and short chain-rule proofs are given in Sec. 2; Appendix C labels supportive feedback-alignment material as intuition rather than a formal theorem. The exact factorization sanity check is cross-referenced in the main text.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The main paper and appendices specify the datasets, architectures, local-rule family, broadcast modes, representative hyperparameters, and compute regime needed to reproduce the headline results, and the supplementary ZIP provides anonymized code, configuration files, and reproduction instructions; see Sec. 4 and Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The anonymous submission includes an anonymized supplementary ZIP containing the core model and training code, selected diagnostic utilities, representative configuration files, tests, and reproduction instructions for the main experimental families. The public benchmarks are cited and accessed through their standard public sources; see Appendix B.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The main text states the benchmark families, seed protocol, representative local-rule settings, and major training choices, while the appendices and supplementary ZIP provide implementation details, configuration files, and operating-mode inventories; see Sec. 4 and Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The paper reports means and standard deviations across 5 seeds for the headline results and across 3 seeds for selected appendix-only support sweeps, states this explicitly in the text, and shows error bars or seed variability in the relevant tables and figures; see Sec. 4 and Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B describes the institutional GPU cluster setting, single A100/H100-class GPUs, approximate memory class, per-family runtime ranges, and approximate GPU-hours for the reported experiments.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses standard public benchmarks and mechanistic simulations, does not involve human-subjects experimentation or restricted data collection, and we are not aware of any aspect that violates the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: The paper is a mechanistic study of dendritic credit assignment using public benchmarks and synthetic diagnostics, with no deployed system, human-subject data, scraped dataset, or direct societal application. The limitations section nevertheless warns against overgeneralizing the mechanism beyond the tested regimes; see Sec. 5.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The submission does not release a high-risk generative model, foundation model, or scraped dataset, so the safeguard question is not applicable here.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.

- 1079 • We recognize that providing effective safeguards is challenging, and many papers do
1080 not require this, but we encourage authors to take this into account and make a best
1081 faith effort.

1082 12. Licenses for existing assets

1083 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
1084 the paper, properly credited and are the license and terms of use explicitly mentioned and
1085 properly respected?

1086 Answer: [\[Yes\]](#)

1087 Justification: The paper credits all benchmark assets, does not redistribute third-party raw
1088 assets, and includes an asset/access table with URLs, citation requirements, and license or
1089 access terms where available; see Appendix B.

1090 Guidelines:

- 1091 • The answer [\[N/A\]](#) means that the paper does not use existing assets.
- 1092 • The authors should cite the original paper that produced the code package or dataset.
- 1093 • The authors should state which version of the asset is used and, if possible, include a
1094 URL.
- 1095 • The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- 1096 • For scraped data from a particular source (e.g., website), the copyright and terms of
1097 service of that source should be provided.
- 1098 • If assets are released, the license, copyright information, and terms of use in the
1099 package should be provided. For popular datasets, paperswithcode.com/datasets
1100 has curated licenses for some datasets. Their licensing guide can help determine the
1101 license of a dataset.
- 1102 • For existing datasets that are re-packaged, both the original license and the license of
1103 the derived asset (if it has changed) should be provided.
- 1104 • If this information is not available online, the authors are encouraged to reach out to
1105 the asset's creators.

1106 13. New assets

1107 Question: Are new assets introduced in the paper well documented and is the documentation
1108 provided alongside the assets?

1109 Answer: [\[Yes\]](#)

1110 Justification: The only new released asset is an anonymized supplementary code ZIP. It
1111 includes an MIT license, README, manifest, reproduction instructions, representative
1112 configuration files, tests, and the core model/training code needed for the main experimental
1113 families; the paper does not release a new dataset, benchmark, or pretrained model.

1114 Guidelines:

- 1115 • The answer [\[N/A\]](#) means that the paper does not release new assets.
- 1116 • Researchers should communicate the details of the dataset/code/model as part of their
1117 submissions via structured templates. This includes details about training, license,
1118 limitations, etc.
- 1119 • The paper should discuss whether and how consent was obtained from people whose
1120 asset is used.
- 1121 • At submission time, remember to anonymize your assets (if applicable). You can either
1122 create an anonymized URL or include an anonymized zip file.

1123 14. Crowdsourcing and research with human subjects

1124 Question: For crowdsourcing experiments and research with human subjects, does the paper
1125 include the full text of instructions given to participants and screenshots, if applicable, as
1126 well as details about compensation (if any)?

1127 Answer: [\[N/A\]](#)

1128 Justification: The work does not involve crowdsourcing or direct research with human
1129 subjects.

1130 Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The work does not involve crowdsourcing or human-subjects research requiring IRB review.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not part of the proposed method, theorem, experiments, or analysis pipeline, so no declaration is required for the core scientific contribution.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.